

# 6 Weeks Summer Internship Program on Machine Learning and Generative Artificial Intelligence using Python Project Guide

## Step 1: Find a Real Problem

Don't start with algorithms.

Start with a problem.

Think about industries such as

| Domain         | Example Problems                         |
|----------------|------------------------------------------|
| Healthcare     | Disease prediction                       |
| Agriculture    | Crop disease detection, yield prediction |
| Education      | Student performance prediction           |
| Finance        | Fraud detection, loan approval           |
| Environment    | Air quality prediction                   |
| Transportation | Traffic prediction                       |
| Retail         | Recommendation systems                   |
| Cybersecurity  | Intrusion detection                      |
| IoT            | Predictive maintenance                   |
| Smart Cities   | Energy consumption prediction            |

## Step 2: Study Existing Research

Students should first understand

- What has already been done?
- What datasets were used?
- Which algorithms worked?
- What are the limitations?

Good websites

- Google Scholar
- Semantic Scholar
- IEEE Xplore
- SpringerLink

- ACM Digital Library
- Elsevier ScienceDirect

What to read

- Review Papers
- Survey Papers
- Recent papers (last 3–5 years)

### Step 3: Find Datasets

#### General AI Datasets

| Website                         | Purpose                    |
|---------------------------------|----------------------------|
| Kaggle                          | ML datasets                |
| UCI Machine Learning Repository | Classical ML               |
| OpenML                          | Benchmark datasets         |
| GitHub                          | Research datasets          |
| Zenodo                          | Research data              |
| Data.gov                        | Government datasets        |
| World Bank Open Data            | Economics                  |
| WHO Data                        | Healthcare                 |
| NASA Open Data                  | Remote sensing             |
| Google Dataset Search           | Search engine for datasets |

### Step 4: Understand the Dataset

Students should answer

- Number of samples
- Features
- Missing values
- Target variable
- Data imbalance
- Labels

### Step 5: Read Papers Properly

For every paper create a table – Atleast 10-15 papers should be studied for Literature review.

| Parameter   | Notes |
|-------------|-------|
| Problem     |       |
| Dataset     |       |
| Algorithm   |       |
| Features    |       |
| Accuracy    |       |
| Limitations |       |
| Future Work |       |

This immediately gives project ideas.

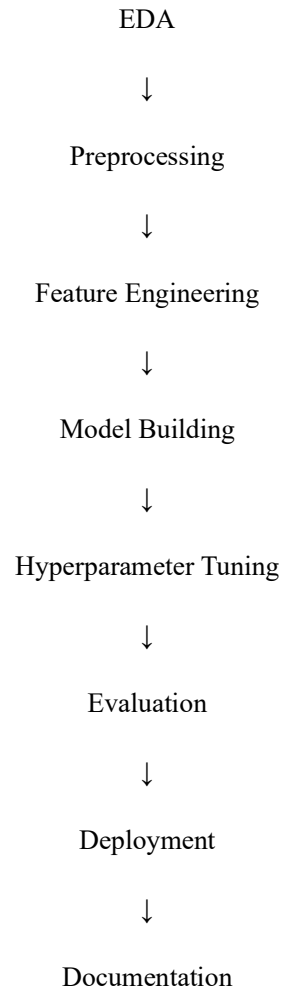
**Step 6: Identify Research Gaps**

| Observation from Existing Research                            | Research Gap                             | Possible Student Project/Improvement                                                                                    |
|---------------------------------------------------------------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Study uses a single dataset or a small dataset.               | Limited generalization and robustness.   | Evaluate the model on multiple or larger datasets and improve performance using data augmentation or transfer learning. |
| Only one algorithm or model is implemented.                   | Lack of comparative analysis.            | Compare multiple machine learning/deep learning models to identify the most effective approach.                         |
| Research focuses only on offline experiments.                 | No real-world deployment or usability.   | Develop a web or mobile application using Streamlit, Flask, or FastAPI for real-time predictions.                       |
| Model provides predictions without explanation.               | Lack of interpretability and user trust. | Integrate Explainable AI (XAI) techniques such as SHAP, LIME, or Grad-CAM to explain predictions.                       |
| Authors mention limitations or future work in the conclusion. | Open research opportunities remain.      | Address one or more of the identified limitations or implement the proposed future work as the project extension.       |

**Step 7: Design Your Own Solution**

Workflow





### **Step 8: Evaluate Properly**

Use metrics like

Classification

- Accuracy
- Precision
- Recall
- F1-score

Regression

- RMSE
- MAE
- $R^2$

## Step 9: Publish

- Research Paper

## Real-World AI Project Ideas

- Heart Disease Prediction
- Diabetes Prediction
- Breast Cancer Detection
- Stroke Prediction
- Chronic Kidney Disease Prediction
- Medical Insurance Cost Prediction
- Hospital Readmission Prediction
- Student Performance Prediction
- Student Dropout Prediction
- Student Placement Prediction
- Loan Approval Prediction
- Credit Card Fraud Detection
- Customer Churn Prediction
- Credit Risk Assessment
- House Price Prediction
- Used Car Price Prediction
- Employee Attrition Prediction
- Employee Salary Prediction
- Employee Performance Prediction
- Customer Segmentation for Retail Businesses
- Product Demand Forecasting
- Sales Forecasting
- Market Basket Analysis
- Customer Purchase Prediction
- Crop Recommendation System
- Crop Yield Prediction
- Fertilizer Recommendation System
- Soil Fertility Classification
- Rainfall Prediction
- Air Quality Prediction
- Water Quality Classification
- Energy Consumption Prediction
- Electricity Load Forecasting
- Traffic Volume Prediction
- Flight Delay Prediction
- Road Accident Severity Prediction
- Spam Email Detection
- Fake News Detection
- Movie Review Sentiment Analysis
- Phishing Website Detection

**Note:** The above topics are provided only as a guide. If you have any other innovative and feasible project ideas related to Machine Learning, you are encouraged to explore them. Feel free to propose your own real-world problem, provided it is approved by the faculty guide/mentor.

## How Students Can Create Their Own Project Ideas

Instead of copying existing projects, follow this simple formula:

1. **Choose a domain** (Healthcare, Agriculture, Finance, Education, etc.).
2. **Find 10–15 recent research papers** on a specific problem.
3. **Create a comparison table** listing datasets, methods, results, and limitations.
4. **Identify a gap**, such as limited data, lack of real-time deployment, or use of only one model.
5. **Formulate your project** by improving one or more aspects (new dataset, better model, multimodal data, explainability, deployment, etc.).
6. **Implement, evaluate, and compare** your approach with the methods from the papers.
7. **Deploy the solution** and document it with code, a report, and a demo.

This approach mirrors how research and industry projects are developed and results in projects that are original, well-motivated, and easier to extend into publications or capstone work.

| Step | Activity                                                        |
|------|-----------------------------------------------------------------|
| 1    | Select domain and define the problem                            |
| 2    | Review 5–10 research papers                                     |
| 3    | Collect and understand the dataset                              |
| 4    | Perform data cleaning and exploratory analysis                  |
| 5–6  | Build baseline models                                           |
| 7    | Improve models and tune hyperparameters                         |
| 8    | Evaluate and compare results                                    |
| 9    | Add advanced features (explainability, multimodal inputs, etc.) |
| 10   | Write documentation and prepare GitHub repository               |
| 11   | Final Research Paper Writing                                    |

## Research Paper Submission Guidelines

- Students must prepare and write the research paper **independently**. The use of **Generative AI tools** for writing or paraphrasing the manuscript is **strictly prohibited**.
- The **AI Similarity Index must be 0%**.

- The **plagiarism similarity index must be less than 10%.**
- Research papers **must not be submitted to or communicated with any conference or journal without the prior approval and supervision of the designated faculty guide.**
- Submission of **both the Research Project and the Research Paper** is **mandatory** for the successful completion of the internship and the issuance of the internship certificate.
- The internship certificate will be issued only upon the successful approval of the project and research paper by the designated Faculty/Guide/Mentor.